# TAI NGUYEN PHU

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## EDUCATION

**University of Information Technology (UIT - VNUHCM)** — **Sep 2022 - Sep 2025**
Bachelor of Computer Science — **GPA:** 3.3/4.0

**Le Khiet High School for the Gifted** — **Sep 2019 - Jun 2022**
Information Technology — **GPA:** 3.4/4.0

## ACHIEVEMENTS

- **IELTS Academic**: 7.0 Overall
- **UIT Global Scholarship**
- **Kaggle Competition: "Credit Risk Model Stability"**: Rank: 1096/3858

## SKILLS

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|---|---|
| **Languages:** | Proficient in Python, C & C++; Basic in Javascript, SQL, HTML, CSS |
| **Deployment Tools:** | Git, GitHub Actions, Docker, AWS, FastAPI, Flask, Weight & Bias |
| **ML/DL Frameworks:** | PyTorch, Transformers, PEFT, Chroma, FAISS, OpenCV, Numpy, Pandas, Scikit-learn, XGBoost |
| **Core AI:** | HuggingFace, LangChain, LangSmith, LangGraph, smolagents |
| **Courseworks:** | Linear Algebra, Discrete Math, Calculus, Statistics and Probability, Data structure and Algorithm, Design and Analysis of Algorithms |

## PROJECTS

**RAG Chatbot with Memory and Multimodal-Multilingual Support** — **May 2025 - July 2025**

- Built a RAG system for scientific PDF files that enables multimodal processing capabilities on text, table, and image data.
- Designed a multilingual, multimodal retrieval pipeline using BGE-M3 embeddings and Chroma vector databases.
- Conducted prompt engineering with multi-turn prompts and scenario-based testing using the Gemini Model API.
- Implemented conversational memory management using LangChain's ConversationSummaryMemory, enabling context-aware follow-up QA in chatbot interactions.
- Implemented a citation-aware abstractive QA mechanism, mapping answers to source documents with text-level references for verifiability.
- Developed an arXiv paper search engine, allowing dynamic retrieval of 150,000+ scientific publications in real-time.
- Containerized and deployed a FastAPI-based end-to-end web application on AWS EC2 and Hugging Face Spaces for reproducibility and scalability.

**Vietnamese Legal Document Semantic Retrieval System** — **Mar 2025 - Apr 2025**

- Fine-tuned a multilingual SBERT model on a curated Vietnamese legal corpus to improve domain-specific retrieval accuracy.
- Achieved NDCG@10: 60.4% and MAP@10: 53.6% on the MTEB benchmark (BKAI Legal Retrieval dataset).
- Built a full semantic retrieval pipeline: data preprocessing, model fine-tuning (contrastive learning), evaluation using MTEB, and deployment.
- Implemented a GPU-accelerated FAISS index with approximately 100K documents for ANN search with sub-second latency.
- Deployed an end-user interface using Gradio, containerized via Docker for scalable and reproducible deployment.

**Glassdoor 2024 Data Science Job Salary Prediction** — **Dec 2024 - Jan 2025**

- Developed a machine learning-powered salary prediction system trained on Glassdoor 2024 Data Science job listings.
- Conducted comprehensive EDA and feature engineering, including data preprocessing, categorical encoding, missing value imputation, and creation of interaction terms to capture complex salary dynamics.
- Achieved $R^2 = 0.815$ with XGBoost, significantly outperforming the baseline Linear Regression ($R^2 = 0.7$).
- Performed hyperparameter optimization via GridSearchCV to maximize model performance.
- Built and deployed an Streamlit web application with real-time salary predictions based on user-provided job attributes.